

VESTA: Evaluating Time-Budgeted Multimodal Briefing for BIST Retail Users

Peri Gunes¹, Ozge Zelal Kucuk², and Harun Benli¹

¹ R&D Center, Infina Software Development Inc., Istanbul, Türkiye
pgunes@infina.com.tr, hbenli@infina.com.tr

² Software Development Faculty, İstanbul Aydın University, Istanbul, Türkiye
ozelalkucuk@stu.aydin.edu.tr

Abstract. Retail investors on Borsa Istanbul (BIST) read KAP list text and charts under a short clock. This paper is an evaluation protocol for that setting, not a new mixer. The public path (i) encodes same-day KAP list teasers as a 16-d bag, (ii) reads a 40-bar candlestick and the OHLCV numbers, and (iii) spends a declared 1-, 3- or 10-minute budget by changing how many tokens are shown while the mixer stays fixed. A 2σ realized-vol flag on the same day’s OHLCV is recovered at 100% F1 by a closed-form rule, so the headline task is next-day BIST100 direction. Score-space mixers and a learned Arevalo GMU (W_v, W_t, W_z on raw 16-d text and the 24×24 image) are reported on public BIST100 data (2018 to 2026). No score-space mixer beats score-space GMU on seed 0 (minimum McNemar $p = 0.46$); learned GMU does not either ($p = 0.29$). Five-seed accuracy sits next to the majority prior (mean fusion 53.0 ± 2.2 versus 52.3). T1/T3/T10 change coverage and templated length, not next-day accuracy (T3 and T10 both 49.7% on seed 0). A long/short overlay loses to buy-and-hold on the 2025 to 2026 bull test window. Human NASA-TLX responses are not claimed. The replication package accompanies the submission.

Keywords: Econometrics · Computational intelligence · Decision support · Multimodal fusion · Borsa Istanbul

1 Introduction

Retail investors sit between the textual density of the Public Disclosure Platform (KAP) [18] and the visual flux of technical charts. The difficulty is not only volume but modality mismatch: a constructive disclosure can coincide with a failed breakout, and the two streams must be reconciled before a decision. Conventional DSS pipelines keep NLP and computer vision in separate pipelines. A second failure mode is temporal: systems emit the same information density whether the user has sixty seconds or ten minutes [7, 17, 22]. On a mobile-first retail platform that mismatch adds intrinsic load [23].

FinMem layers trading-agent memory by information half-life [27]. VESTA scores a briefing whose clock is 1, 3 or 10 minutes: KAP list text and a 40-bar

chart. The fusion rule does not change with the clock; only how many tokens are shown. T3 and T10 attach an M3 day-level hop; that analogue does not enter the mixer. Decision support is scored on next-day BIST direction, not on a realized-volatility flag that can be read off the same day’s OHLCV table. Whenever a chart encoder is used, a numeric OHLCV baseline is reported beside it [11]. The public tables mix unimodal class scores with GMU [2], tensor fusion [28] and a scalar gate, and they add one learned-GMU row that fits Arevalo’s W_v, W_t, W_z on the raw vectors. Actionability is a long/short overlay against buy-and-hold. The public chart is bars $t - 40, \dots, t - 1$. We do not claim a human-load result: a 12-participant NASA-TLX protocol is specified in `study/`; responses are not collected.

The contribution is four protocol facts, not a new mixer. (1) T1, T3 and T10 change how many tokens are shown while the mixer stays fixed. (2) A 2σ realized-vol flag on the same day’s OHLCV is recovered at 100% F1 by a closed-form rule, so the headline label is next-day direction. (3) Tiers measure delivery (coverage, declared latency, tokens, length-normalized noise); they do not raise next-day accuracy. Action is a long/short overlay against buy-and-hold on one bull test window. (4) Score-space mixers are not separable from score-space GMU on seed 0 (minimum McNemar $p = 0.46$); learned GMU is not either ($p = 0.29$). We do not name a winner [10]. Those four facts are in the tables. The objects match three CoMeSySo areas: an econometric next-day index label, computational-intelligence mixers, and software whose delivered length depends on a declared time budget.

1.1 Related work

Financial NLP has moved from sentiment scoring [20] to domain LLMs such as FinGPT [25]. Hallucination on prices remains a documented failure mode [13], so public text here is KAP list teasers, not free generation [15]. Tool-using agents [26] and FinMem [27] are the closest named relatives; this paper scores a briefing, not a trading agent. The public hop is M3 day-level cosine, not a portfolio retrieve-then-generate loop.

1.2 Multimodal fusion and chart understanding

GMU learns a multiplicative gate between modalities [2]; Tensor Fusion Networks use an outer product [28]; MulT uses directional pairwise cross-modal attention [24]. Jiang and Ji add a forget-gate over cross-modal attention maps [12]. Public tables mix unimodal class scores with GMU, TFN, a scalar gate, mean, concat and one CPU attention layer. That layer is not the MulT of Tsai et al.: projections are frozen and there is a single head. A learned-GMU row fits Arevalo’s maps on the raw vectors.

Pix2Struct and MatCha pretrain screenshot-to-text models [14, 16]. Jiang, Kelly and Xiu show that CNNs on price-trend images extract return-predictive patterns [11]. That result requires a tabular OHLCV baseline beside any chart

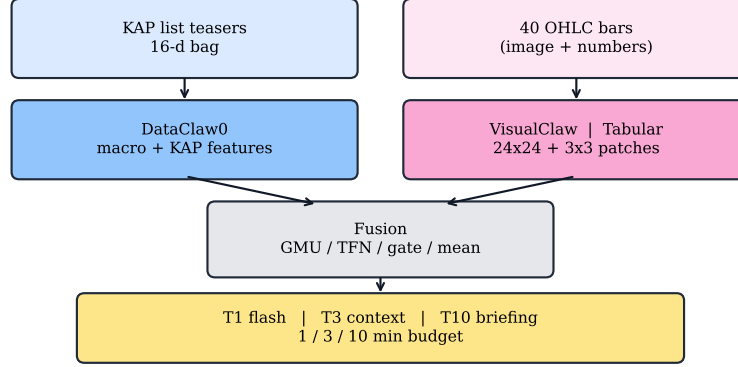


Fig. 1. Public path: KAP list bag and a 40-bar chart go through a mixer; T1, T3 or T10 spends the 1-/3-/10-minute budget.

encoder. Akturk et al. forecast BIST100 from numeric indicators [1]; those features are the control whenever a chart encoder is scored.

1.3 Cognitive load

Presented density should match working-memory limits [22, 23, 17, 7]. Adaptive interfaces are judged by the task, not by whether a widget changed [9, 8]. Length-blind summarization metrics reward brevity [21], which is why Table 6 reports IN_{new} on delivered tokens. That table is a payload count. It is not a human-load result. The NASA-TLX protocol is Sect. 2.6; this manuscript has no responses.

2 Methods

Methods cover the pipeline, the mixers, time-budgeted delivery, and the evaluation design. VESTA has three layers (Fig. 1). Perception uses a text module and a chart module; an off-the-shelf block mixes them; a temporal layer spends the user’s time budget.

2.1 DataClaw0: text brief

The public mixer path is a 16-d macro+KAP bag plus a templated brief. The retrieve-score hop is executed on cached BGE-M3 KAP-day vectors [3, 15, 26]. $\alpha = 1$. The β gold / USD/TRY / session terms are not fitted and are not used. There is no user portfolio.

For each test day the hop is the prior KAP day with the highest cosine to that day’s M3 vector. A same-date XU100 constituent with a KAP teaser is recorded as a peer hop. The retrieved analogue does not enter the 16-d mixer.

Table 1. Example KAP list teasers in VESTA-Public (subject plus short summary from kap.org.tr). KAP pol. is kap_polarity, the KAP-only silver codebook, not mixed text_polarity.

Date	Ticker	KAP list teaser	KAP pol.
2018-06-14	BIMAS	Payların Geri Alınmasına İlişkin Bildirim. 13	bullish
		Haziran 2018 Tarihli Pay Geri Alım İşlemleri	
2018-06-19	ASELS	BISTECH Pay Piyasası Alım Satım Sistemi	bullish
		Duyurusu. Temettü Ödemesi	
2018-05-25	YKBNK	Özel Durum Açıklaması (Genel). Bankamıza	bearish
		verilen idari para cezasının iptaline ilişkin	
		açılan dava hk.	
2020-08-11	FROTO	Özel Durum Açıklaması (Genel). Rekabet	bearish
		Kurulu Soruşturma Kararı Hakkında	

T3 and T10 still spend a `hist_analog` token; the console attaches the retrieved date and teaser.

The text dataset is VESTA-Public: one row per ticker-day (37,046 events; 27 BIST names plus XU100; 24 May 2018 to 19 August 2026). A row is kept if the name moved at least 2%, the vol flag fired, RSI was extreme, or a KAP list item was published that calendar day. 19,645 rows have a same-day KAP teaser; 15,212 have a cached public HTML body. Yahoo headlines exist for 41 rows. Table 1 shows four KAP list strings. We do three things with them: (i) a Turkish/English lexicon plus subject priors (dividend and buyback tilt bullish; fine and probe tilt bearish) yields `kap_polarity`; (ii) the same strings, plus USD/TRY, gold and the session, become an eight-plus-eight feature vector for the mixers; (iii) MiniLM [19] and BGE-M3 embed the concatenated list text. The silver event-row brief for the first Table 1 day, rebuilt from the stored columns, is: “2018-06-14 BIMAS.IS. Session return +6.33%. USD/TRY +1.65%, gold +0.55%. Ann. realized vol 33.8%, RSI 48. Brief polarity bullish; chart cue none. Payların Geri Alınmasına İlişkin Bildirim. 13 Haziran 2018 Tarihli Pay Geri Alım İşlemleri”. That string is the ticker-day record, not the T10 token bag.

2.2 VisualClaw: candles, pixels, tokens

VisualClaw draws the chart from OHLC bars. Figure 2 is one XU100 window from the test split.

1. Take the 40 sessions strictly before day t (open, high, low, close). Today’s candle is not in the picture.
2. Paint a grayscale candlestick: a one-pixel wick and a rectangular body. Close $\geq$ open is dark (30/255); close $<$ open is light (180/255). Price is min-max scaled to that window.
3. Resize to 24×24 . The vision MLP flattens this to a 576-d vector and classifies with hidden layers 64-32. That is the vision row in Table 4.

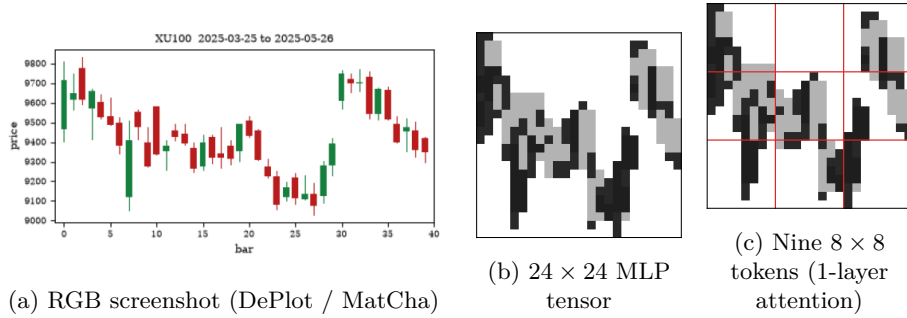


Fig. 2. VisualClaw on one XU100 test window (40 bars before 27 May 2025). (a) RGB screenshot for DePlot / MatCha. (b) 24×24 grayscale tensor the MLP flattens. (c) Nine 8×8 tokens for the 1-layer attention mixer. That mixer is not Tsai Mult.

4. For the 1-layer attention mixer only, cut the same 24×24 image into a 3×3 grid of 8×8 patches (nine visual tokens). The 16-d text brief is split into two tokens. Projections are frozen (seed 0); only the MLP on top is trained. That mixer is not Tsai Mult.

Two extra probes sit beside the mixer: a frozen ImageNet ViT-B/16 [6] that resizes a colour chart to 224×224 and uses that model’s own 16×16 patches; and zero-shot DePlot / MatCha, which see a matplotlib RGB screenshot of the same 40 bars (Fig. 2(a)). Tabular OHLCV at day t is always reported beside the picture [11]. Public tables train the flattened 24×24 MLP; they do not add a recency term to patch positions.

2.3 Fusion

Arevalo et al. write the bimodal GMU as [2]

$$h_v = \tanh(W_v x_v), \quad h_t = \tanh(W_t x_t), \quad (1)$$

$$z = \sigma(W_z [x_v; x_t]), \quad h = z h_v + (1 - z) h_t, \quad (2)$$

so z near 1 keeps the visual branch. Zadeh et al. flatten $[h_t; 1] \otimes [h_v; 1]$ (TFN) [28]. Table 4 has two GMU rows. The score-space GMU takes unimodal class-score vectors as x_t and x_v and sets z from the score gap (Arevalo’s mix; W_v, W_t, W_z unused in that row only). The learned-GMU row fits those three maps on the raw 16-d text vector and the flattened 24×24 image by minibatch gradient descent, then a linear head on h . The TFN row uses the outer product on the same class scores. The scalar-gate row uses

$$g = \sigma(4(p_{\text{text}} - p_{\text{vision}})), \quad (3)$$

$$h = g \tanh(p_{\text{text}}) + (1 - g) \tanh(p_{\text{vision}}), \quad (4)$$

so g near 1 keeps text. A small MLP reads the mix. Mean and concat are unweighted. The 1-layer attention row is one CPU attention layer on two text

tokens and nine 8×8 patches, with frozen projections. That layer is not Tsai Mult [24]. No mixer is named as the winner. The fused call drives T1 / T3 / T10.

2.4 Temporal Orchestration Layer

The intended independent variable is the user’s clock $B \in \{1, 3, 10\}$ minutes. No participant sat that cap. Table 6 reports declared latencies 0.9 / 2.1 / 3.4s, not wall-clock. T3 and T10 emit the same fused call; extra tokens are templates.

- T1 (flash). Emit a gated alert only if $|g - 0.5| > \tau$ with $\tau = 0.18$; otherwise abstain. Declared latency 0.9s.
- T3 (context). Always emit the fused call plus a retrieved analogue token (M3 cosine, $\alpha = 1$; no portfolio).
- T10 (briefing). Expand into a macro explanation and a hedge sketch (e.g. airline exposure versus a gold certificate). The 3.4s figure in Table 6 is a declared budget, not measured wall-clock.

The mixer stays fixed. Tiers change how many tokens are shown. Section 3.5 reports coverage, latency, tokens and noise per tier.

2.5 Evaluation protocol

The evaluation has a leakage diagnostic, a primary next-day direction task, per-tier delivery metrics, and a paper-trading overlay.

Diagnostic: the realized-volatility label The 2σ realized-vol flag at day t is 20-day realized vol (annualized) above the 60-day rolling mean plus 2σ of that 20-day vol series, all taken from the tabular vector at t (100% F1). The public chart does not contain today’s candle, so vision F1 on this label is lag-1 recovery. We report the flag only as a diagnostic.

Primary task: forward direction The primary label at close of day t is the next session’s direction: $y_{t+1} = 1$ if the BIST100 simple return r_{t+1} is positive, and $y_{t+1} = 0$ otherwise. The label is unknown at the close of day t . We also report a five-session horizon. Splits are chronological (70/15/15) to block look-ahead [1].

Metrics Classification. Accuracy and macro-F1, mean ± 1.96 standard error over five seeds. McNemar with continuity correction [5] on seed 0 of one chronological holdout. Table 4 has fourteen rows. Pairwise McNemar versus score-space GMU is a control, not a winner (minimum $p = 0.46$ for score-space mixers; $p = 0.29$ for learned GMU). With that many rows we follow Harvey et al. [10] and do not pick a winner after an uncorrected search.

Sentiment / technical accuracy. Table 2 in Sect. 3 reports proxy accuracy versus next-day direction. Macro flags and KAP polarity are deterministic rules on one chronological split (no seed CI). Vision and tabular are five-seed means from Table 4. Three KAP codebooks on the 10k slice give Fleiss $\kappa = 0.50$; there are no human annotators. Chart A/B Cohen $\kappa = 0.52$ [4]. The 250-row sample has no annotators; those columns are codebook B.

Information noise. This is the computational cognitive measure. Let N_{raw} be tokens in the unprocessed window (the raw KAP list plus session flags) and N_{del} tokens actually delivered in the T1, T3 or T10 briefing. Let N_{irrel} be delivered tokens that are not in a relevance lexicon for that window (session, USD/TRY, gold, vol, KAP digest). The index $\text{IN}_{\text{old}} = N_{\text{irrel}}/N_{\text{raw}}$ rewards short answers: a 40-token briefing that is 80% filler still looks clean against a 700-token feed [21]. We therefore report

$$\text{IN}_{\text{new}} = N_{\text{irrel}}/N_{\text{del}}, \quad (5)$$

the fraction of the delivered finance briefing that is non-actionable, together with the compression ratio $N_{\text{del}}/N_{\text{raw}}$. IN_{new} is our index; Sun et al. motivate dividing by delivered length. Both counts use templated token bags with filler padding, so IN_{new} rises by construction when T3/T10 add filler. Table 6 is that computational proxy; human NASA-TLX is unspecified (Sect. 2.6).

Decision quality. Classification F1 does not say whether acting on the output helps [8]. We paper-trade a long/short overlay: +1 if the model predicts up, −1 otherwise, with 5 bp round-trip cost, and compare Sharpe, total return and max drawdown to buy-and-hold. Table 7 is that control; Sect. 2.6 is the human cognitive protocol.

Public replication corpus Numbers below use VESTA-Public: XU100.IS, USDTRY and COMEX gold from 24 May 2018 to 19 August 2026 (2,052 index windows after a 40-day lookback; test $n = 308$ from 27 May 2025), 27 BIST names, and the public KAP list inventory of 39,890 filings (27 names; scrape window 2 January 2018 to 27 August 2026). Tables 2 to 7 are index-level; the 37,046-row event panel also labels constituents. KAP-only polarity: 7,172 / 3,742 / 26,132 (bullish / bearish / neutral); 15,212 rows have a cached public HTML body. Mixers use the 16-d macro+KAP vector; MiniLM and BGE-M3 are uni-modal text probes on the same list teasers.

2.6 Human protocol

The cognitive work has two layers. The first is computational and is reported with the tier table in Sect. 3.5. A BIST retail user on a mobile desk has about one, three or ten minutes between a KAP drop and an order. That clock is the independent variable. The dependent quantities are coverage (how often the system speaks), declared latency, tokens delivered, and IN_{new} (what fraction of the delivered briefing is non-actionable). The token bags are built from the day’s finance flags: session return, USD/TRY, gold, realized vol, a KAP digest, and,

Table 2. Proxy accuracy versus next-day direction y_{t+1} (test $n = 308$). Macro/KAP are single-split rules; vision/tabular are mean ± 1.96 se over five seeds.

Proxy	Acc. (%)
Macro flags (USD/TRY, gold, vol, shock)	54.9
KAP polarity (lexicon on public list text)	49.7
Vision 24×24 MLP	52.9 ± 2.3
Tabular OHLCV	50.6 ± 0.0

on T3/T10, a historical analogue and a hedge sketch. Those counts measure load on the payload. They do not measure load on a person [9, 8].

The second layer is a human protocol in **study/**. Twelve retail-account holders, within-subject, would see eight sealed snapshots from the public test window (two XU100 sessions and six constituents: SASA, SISE, YKBNK, TAVHL, AKBNK, PETKM). The sealed gold is the subsequent week’s realized direction of that name. Conditions are the raw KAP+OHLC feed, a unimodal text or chart, or the matching VESTA tier, under a 1-, 3- or 10-minute cap. Outcomes are the week-direction call, time-to-decision, NASA-TLX, and a 7-point trust item. Consent and the bilingual instrument are in **study/**.

Responses have not been collected ($n = 0$). This paper does not claim a human-load or trust result. A model dry-run on the sealed gold hits 4/8 name-level weeks for the raw session sign; index mixers are scored only on the two XU100 weeks (text 0/2, gated 1/2). NASA-TLX and trust stay blank. The executable protocol remains in **study/** so a later sample can be scored against the same gold.

3 Results

Unimodal encoders are sklearn MLPs (64-32, Adam, early stopping) on standardized features. Vision sees a 24×24 grayscale candlestick of bars $t - 40, \dots, t - 1$; tabular sees eleven OHLCV/macro features at t , including the ingredients of the leaked vol rule; text sees the sixteen-dimensional macro+KAP vector. Five seeds $\{0, \dots, 4\}$. Training is on CPU. Public BGE-M3 and frozen ViT-B/16 are unimodal probes; they remain unimodal in Table 4.

3.1 Proxy accuracy

Table 2 is the rule and encoder proxy against next-day direction. Macro flags are the strongest single-split rule (54.9%). KAP polarity on list teasers is 49.7%. The 24×24 vision MLP and tabular OHLCV repeat their five-seed means from Table 4.

Table 3. Diagnostic task: 2σ realized-vol label, computable from the tabular vector at t . The closed-form row is a single split. MLP rows are mean ± 1.96 se over five seeds. Vision sees bars $t - 40, \dots, t - 1$ only.

Method	Accuracy (%)	Macro-F1 (%)
Closed-form OHLCV rule	100.0	100.0
Tabular MLP (same numbers)	96.8 ± 0.0	86.6 ± 0.0
Vision MLP (chart image)	90.6 ± 1.1	58.2 ± 3.0

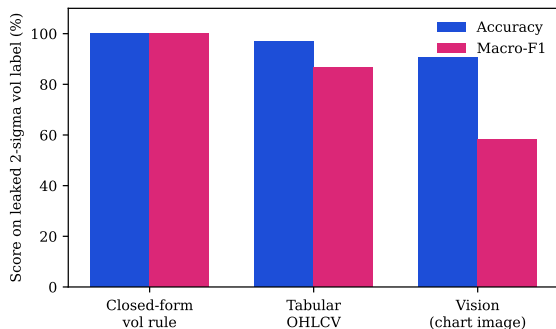


Fig. 3. Diagnostic 2σ -vol label. The closed-form OHLCV rule scores 100% F1; the chart omits day t .

3.2 Diagnostic: the leaked volatility label

Table 3 and Fig. 3 confirm the closed-form critique. The 2σ rule on the tabular vector scores 100% F1. A tabular MLP that is given those numbers reaches 96.8% accuracy (macro-F1 86.6%; the F1 drop is class imbalance: 25 positives among 308 test days, 8.1%; 9.4% of all 2,052 windows). The 24×24 vision encoder, which never sees day t 's candle, falls to 90.6% accuracy and 58.2% F1.

3.3 Primary task: next-day direction

The next-day signal is null. Five-seed accuracy CIs overlap the majority prior 52.3%. Mean fusion 53.0 ± 2.2 is not a winner. Table 4 reports the leakage-free task (test prior 52.3% up-days). Accuracy / macro-F1: text-only (macro+KAP) 52.1 ± 2.4 / 47.3 ± 1.5 against majority 52.3 / 34.3; MiniLM 53.0 ± 0.4 / 50.7 ± 1.9 ; public BGE-M3 51.1 ± 1.9 / 50.2 ± 2.1 ; frozen ViT-B/16 50.9 ± 1.4 / 50.3 ± 1.6 . Mixers share 16-d text and the 24×24 image. Among score-space mixers, none beats score-space GMU on seed 0 (minimum McNemar $p = 0.46$). Learned GMU (W_v, W_t, W_z on the raw vectors) is 52.5 ± 0.5 / 40.3 ± 4.4 and does not beat score-space GMU either (seed-0 McNemar $p = 0.29$); the F1 drop is majority collapse. On the five-session horizon (seed 0) tabular accuracy / F1 is 59.1% / 39.9%; vision is 51.9% / 49.0% and the scalar gate is 50.3% / 49.3%.

Table 4. Next-day BIST100 direction on VESTA-Public (test $n = 308$). Mean $\pm$ 1.96 se, five seeds. Rows are ordered by macro-F1. MiniLM, BGE-M3 and ViT are unimodal probes. Mixers include 1-layer attention (not MulT), GMU [2] and TFN [28]. Score-space GMU/TFN/gate mix unimodal class scores. Learned GMU fits Arevalo’s W_v, W_t, W_z on raw text and the 24×24 image. No mixer is named as the winner.

Configuration	Accuracy (%)	Macro-F1 (%)
Mean fusion	53.0 ± 2.2	52.4 ± 2.3
TFN	52.1 ± 2.1	51.7 ± 2.1
Vision-only (24×24 MLP)	52.9 ± 2.3	51.4 ± 2.6
Scalar gate (score-space)	51.8 ± 1.7	51.4 ± 1.8
Concatenation	52.8 ± 1.8	51.1 ± 2.3
Text-only (MiniLM KAP)	53.0 ± 0.4	50.7 ± 1.9
GMU (score-space)	50.6 ± 2.4	50.6 ± 2.4
Vision-only (frozen ViT-B/16)	50.9 ± 1.4	50.3 ± 1.6
Text-only (public BGE-M3)	51.1 ± 1.9	50.2 ± 2.1
1-layer attention (not MulT)	50.2 ± 3.1	48.4 ± 3.0
Text-only (macro + KAP)	52.1 ± 2.4	47.3 ± 1.5
Tabular OHLCV	50.6 ± 0.0	41.8 ± 0.0
Learned GMU (W_v, W_t, W_z)	52.5 ± 0.5	40.3 ± 4.4
Majority class	52.3 ± 0.0	34.3 ± 0.0

Table 5. Zero-shot CPU screenshot VLMs on the full test window ($n = 308$), with no fine-tune.

Method	Acc. vs y_{t+1}	Macro-F1	Acc. vs window trend
Majority	52.3	34.3	n/a
Numeric window trend	51.6	49.5	100.0
DePlot (Pix2Struct)	52.9	51.9	77.9
MatCha-ChartQA	52.3	46.0	58.4

Replacing the gate with a fixed mean changes five-seed F1 from 51.4 ± 1.8 to 52.4 ± 2.3 (seed-0 McNemar $p = 1.00$).

3.4 Zero-shot Pix2Struct and MatCha

Zero-shot DePlot [14] and MatCha-ChartQA [16] run on CPU matplotlib screenshots of the full test window ($n = 308$), with no fine-tune. Calls are $\text{sign}(\hat{y}_{\text{last}} - \hat{y}_{\text{first}})$. Table 5 also scores the numeric window trend of the input OHLC, which is still not y_{t+1} . DePlot parses 307/308 images and recovers that trend on 77.9% of days; next-day accuracy is 52.9%. MatCha-ChartQA often returns the bar index (parse rate 166/308) and matches the majority class on y_{t+1} .

3.5 Per-tier measurement

Table 6 and Fig. 5 measure the Temporal Orchestration Layer. Latencies 0.9 / 2.1 / 3.4 s are declared budgets, not wall-clock. No participant sat the 1-/3-

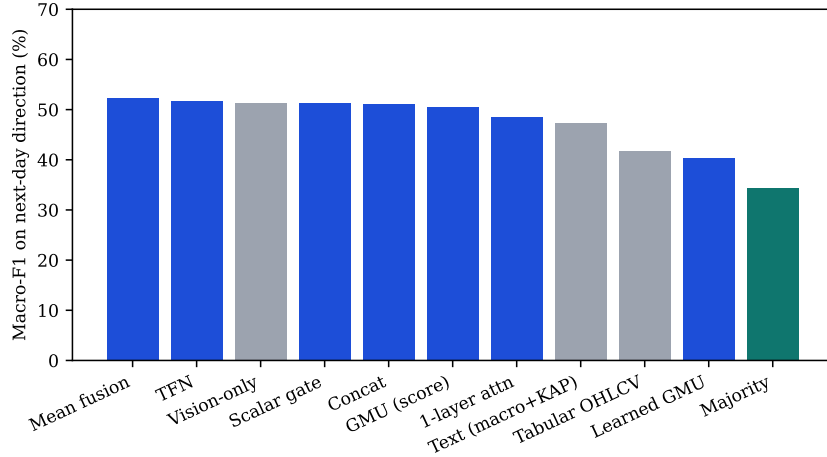


Fig. 4. Macro-F1 for score-space mixers, sklearn unimodal rows, and learned GMU from Table 4. MiniLM, BGE-M3 and ViT stay in the table only. Mean $\pm$ 1.96 se is in the table.

Table 6. Temporal Orchestration Layer on seed 0. Latency is a declared budget, not measured GPU time. Acc. is among emitted calls; T1 overall accuracy with majority fill is 55.2%.

Tier	Cov. (%)	Lat. (s)	Acc. (%)	Tokens	IN _{old}	IN _{new}
T1 flash	32.5	0.9	60.0	6	0.5%	60.3%
T3 context	100.0	2.1	49.7	17	1.9%	82.1%
T10 briefing	100.0	3.4	49.7	52	6.8%	93.6%

/10-minute cap. On seed 0, T1 covers 32.5% of days (60.0% among emitted calls; 55.2% after majority fill). T3 and T10 always emit (coverage 100%). T3 accuracy equals T10 accuracy (both 49.7%) because extra tokens are templates, not new evidence. The M3 hop retrieves on all 308 test days (mean cosine 0.89). The retrieved day’s next-day sign matches y_{t+1} on 49.8% of 305 scored days. That analogue does not beat the majority prior, and it does not enter the mixer. $IN_{old} < 7\%$ for every tier. IN_{new} is 60.3 / 82.1 / 93.6%; that series is the padding index. It rises by construction when T3/T10 add filler.

3.6 Decision quality

Table 7 asks whether acting on the output would have helped. In the May 2025 to August 2026 test window, BIST100 buy-and-hold returned +56.3% (Sharpe 1.69). Every seed-0 overlay we tried underperformed that long-only benchmark after 5 bp costs. Vision-only (Sharpe 0.37) and the scalar gate (0.33) were positive

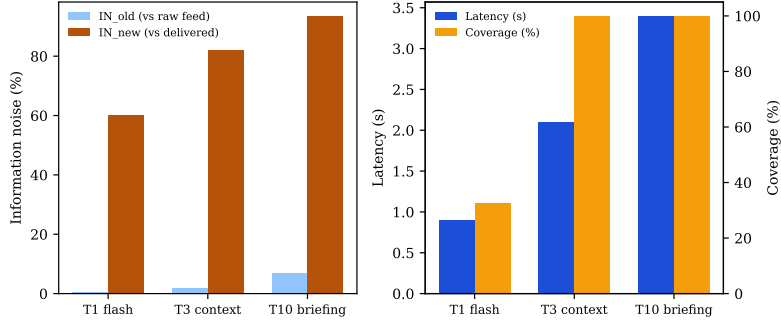


Fig. 5. Left: length-normalized noise of the delivered briefing (IN_{old} versus IN_{new}). Right: declared latency and coverage for T1, T3 and T10.

Table 7. Paper-trading overlay on seed-0 calls, one bull chronological window (27 May 2025 to 19 August 2026), 5 bp costs. BH is long-only BIST100. Dir. is the fraction of days the signed position matches y_{t+1} ; for buy-and-hold that equals the up-day prior. Long/short versus BH is a hard test on this window; there is no second market.

Policy	Dir. (%)	Sharpe	Total ret.	Max DD
Buy-and-hold	52.3	1.69	+56.3%	14.5%
Vision-only overlay	50.0	0.37	+7.6%	16.4%
Scalar-gate overlay	49.7	0.33	+6.3%	17.1%
Tabular overlay	50.6	-0.11	-6.2%	23.0%
GMU overlay	48.1	-0.74	-21.8%	27.4%

but far below BH; GMU and tabular were negative. A bull-market holdout is a hard test for a long/short overlay [10].

4 Discussions

This paper is a protocol, not a mixer. The public path runs an M3 day-level hop ($\alpha = 1$, no portfolio); the analogue hit rate is 49.8% and the hop does not enter the mixer. Tiers change delivery, not next-day accuracy: T3 and T10 both score 49.7% on seed 0 because extra tokens are templates. Five-seed accuracy sits next to the majority prior; we do not name a winner. The overlay is one bull window; it loses to buy-and-hold. Human NASA-TLX is unspecified ($n = 0$). Fleiss $\kappa = 0.50$ is three codebooks; there are no human annotators. Public text is KAP list teasers, not full KAP HTML.

Learned GMU does not beat score-space GMU ($p = 0.29$). There is no score-space mixer that beats score-space GMU either (minimum $p = 0.46$). The learned-GMU row is the control for fusing unimodal class scores. The public chart uses bars $t - 40, \dots, t - 1$. Information noise and tier latencies are simulated. IN_{new} rises by construction when T3/T10 add filler.

Three items remain open. The 12-participant NASA-TLX study in `study/` has no responses, so human load and trust are unspecified. There is no human-annotator gold set; polarity is silver codebook output. Full disclosure-detail pages and a second market are absent. Those are the next measurements.

Disclosure of Interests. The authors declare that they have no competing interests relevant to the content of this article. P.G. and H.B. are employed by Infina Software Development Inc., which operates a BIST retail platform; the public replication does not use proprietary KAP text.

References

1. Akturk, Z., Kılıç, E., Algorabi, Ö., Ulu, M., Türkan, Y.S., Namlı, E.: Real-time forecast of BIST100 index under market volatility and uncertainty. *Computational Economics* **67**(4), 3249–3274 (2026). <https://doi.org/10.1007/s10614-025-10986-1>
2. Arevalo, J., Solorio, T., y Gómez, M.M., González, F.A.: Gated multi-modal units for information fusion. *arXiv preprint arXiv:1702.01992* (2017). <https://doi.org/10.48550/arXiv.1702.01992>
3. Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., Liu, Z.: M3-embedding: Multilinguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In: *Findings of the Association for Computational Linguistics: ACL 2024*. pp. 2318–2335 (2024). <https://doi.org/10.18653/v1/2024.findings-acl.137>
4. Cohen, J.: A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* **20**(1), 37–46 (1960). <https://doi.org/10.1177/001316446002000104>
5. Dietterich, T.G.: Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation* **10**(7), 1895–1923 (1998). <https://doi.org/10.1162/089976698300017197>
6. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N.: An image is worth 16x16 words: Transformers for image recognition at scale. In: *International Conference on Learning Representations* (2021), [arXiv:2010.11929](https://arxiv.org/abs/2010.11929)
7. Eppler, M.J., Mengis, J.: The concept of information overload: A review of literature from organization science, accounting, marketing, MIS, and related disciplines. *The Information Society* **20**(5), 325–344 (2004). <https://doi.org/10.1080/01972240490507974>
8. Findlater, L., Gajos, K.Z.: Design space and evaluation challenges of adaptive graphical user interfaces. *AI Magazine* **30**(4), 68–73 (2009). <https://doi.org/10.1609/aimag.v30i4.2268>
9. Gajos, K.Z., Czerwinski, M., Tan, D.S., Weld, D.S.: Exploring the design space for adaptive graphical user interfaces. In: *Proceedings of the Working Conference on Advanced Visual Interfaces (AVI '06)*. pp. 201–208 (2006). <https://doi.org/10.1145/1133265.1133306>
10. Harvey, C.R., Liu, Y., Zhu, H.: ... and the cross-section of expected returns. *The Review of Financial Studies* **29**(1), 5–68 (2016). <https://doi.org/10.1093/rfs/hhv059>

11. Jiang, J., Kelly, B., Xiu, D.: (Re-)Imag(in)ing price trends. *The Journal of Finance* **78**(6), 3193–3249 (2023). <https://doi.org/10.1111/jofi.13268>
12. Jiang, M., Ji, S.: Cross-modality gated attention fusion for multi-modal sentiment analysis. *arXiv preprint arXiv:2208.11893* (2022). <https://doi.org/10.48550/arXiv.2208.11893>
13. Kang, H., Liu, X.Y.: Deficiency of large language models in finance: An empirical examination of hallucination. In: *I Can’t Believe It’s Not Better Workshop: Failure Modes in the Age of Foundation Models* (2023), *neurIPS 2023 workshop*. <https://openreview.net/forum?id=SGiQxu8zFL>. *arXiv:2311.15548*
14. Lee, K., Joshi, M., Turc, I.R., Hu, H., Liu, F., Eisenschlos, J.M., Khandelwal, U., Shaw, P., Chang, M.W., Toutanova, K.: Pix2Struct: Screenshot parsing as pretraining for visual language understanding. In: *Proceedings of the 40th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 202, pp. 18893–18912 (2023)
15. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., tau Yih, W., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: *Advances in Neural Information Processing Systems*. vol. 33, pp. 9459–9474 (2020)
16. Liu, F., Piccinno, F., Krichene, S., Pang, C., Lee, K., Joshi, M., Altun, Y., Collier, N., Eisenschlos, J.: MatCha: Enhancing visual language pretraining with math reasoning and chart derendering. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. pp. 12756–12770 (2023). <https://doi.org/10.18653/v1/2023.acl-long.714>
17. Miller, G.A.: The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review* **63**(2), 81–97 (1956). <https://doi.org/10.1037/h0043158>
18. Public Disclosure Platform (KAP): KAP public disclosure platform. <https://www.kap.org.tr> (2026), accessed 27 August 2026
19. Reimers, N., Gurevych, I.: Sentence-BERT: Sentence embeddings using siamese BERT-networks. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. pp. 3982–3992 (2019). <https://doi.org/10.18653/v1/D19-1410>
20. Sohangir, S., Wang, D., Pomeranets, A., Khoshgoftaar, T.M.: Big data: Deep learning for financial sentiment analysis. *Journal of Big Data* **5**(1), 3 (2018). <https://doi.org/10.1186/s40537-017-0111-6>
21. Sun, S., Shapira, O., Dagan, I., Nenkova, A.: How to compare summarizers without target length? pitfalls, solutions and re-examination of the neural summarization literature. In: *Proceedings of the Workshop on Methods for Optimizing and Evaluating Neural Language Generation*. pp. 21–29 (2019). <https://doi.org/10.18653/v1/W19-2303>
22. Sweller, J.: Cognitive load during problem solving: Effects on learning. *Cognitive Science* **12**(2), 257–285 (1988). https://doi.org/10.1207/s15516709cog1202_4
23. Sweller, J.: The development of cognitive load theory: Replication crises and incorporation of other theories can lead to theory expansion. *Educational Psychology Review* **35**(4), 95 (2023). <https://doi.org/10.1007/s10648-023-09817-2>
24. Tsai, Y.H.H., Bai, S., Liang, P.P., Kolter, J.Z., Morency, L.P., Salakhutdinov, R.: Multimodal transformer for unaligned multimodal language sequences. In: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. pp. 6558–6569 (2019). <https://doi.org/10.18653/v1/P19-1656>

25. Yang, H., Liu, X.Y., Wang, C.D.: FinGPT: Open-source financial large language models. arXiv preprint arXiv:2306.06031 (2023). <https://doi.org/10.48550/arXiv.2306.06031>
26. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y.: ReAct: Synergizing reasoning and acting in language models. In: International Conference on Learning Representations (2023), arXiv:2210.03629
27. Yu, Y., Li, H., Chen, Z., Jiang, Y., Li, Y., Zhang, D., Liu, R., Suchow, J.W., Khashanah, K.: FinMem: A performance-enhanced LLM trading agent with layered memory and character design. Proceedings of the AAAI Symposium Series **3**(1), 595–597 (2024). <https://doi.org/10.1609/aaais.v3i1.31290>
28. Zadeh, A., Chen, M., Poria, S., Cambria, E., Morency, L.P.: Tensor fusion network for multimodal sentiment analysis. In: Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing. pp. 1103–1114 (2017). <https://doi.org/10.18653/v1/D17-1115>